SYSTEMS ANALYSIS AND DESIGN | GROUP PROJECT

School Gate Entry Management System

Student Task Sheet: Analyse, Design and Prototype

Project duration: 9 weeks | Group size: 4 to 5 students | Total marks: 100

Group name:		**Class / Stream:**	
Members:		**Submission date:**	

1. The Scenario

Your school currently controls its gate using a paper visitor book and the guard's memory. The book is slow to fill in, often illegible, and easily lost. Nobody can say with confidence who is on the school compound at a given moment, which is a serious problem during an emergency or fire drill. Late arrivals are not reliably recorded, students can leave early without proper approval, and parents are rarely told when something happens.

The principal has asked your team, acting as a **systems analysis and design consultancy**, to study the current gate procedure and to design a computerised system that makes entry and exit **secure, fast and well recorded**.

2. Your Task

As a team you must:

- Investigate the current gate system and identify its problems.
- Gather real evidence from the people who use it (guard, deputy principal, teachers, students).
- Specify what the new system must do and how well it must do it.
- Model the system with standard analysis and design diagrams.
- Design the database, input forms, reports and security controls.
- Build and test a working prototype (see Phase 6 for allowed tools).
- Present your final solution to the "client" (your teacher and invited staff).

3. What the System Should Cover

Your design must include the **core features**. Choose **at least two** optional features.

	Feature	Description
Core	Student check-in / out	Record every entry and exit with date and time, by ID card (barcode/QR) or manual search.
Core	Late arrival log	Automatically flag students who arrive after the cut-off time (e.g. 7:30 am) and record a reason.

	Feature	Description
Core	Early exit permission	A teacher or administrator approves; the guard sees the approval before letting the student out.
Core	Visitor registration	Name, ID number, purpose, person visited, badge number, time in and out.
Core	User roles and login	Guard, administrator, teacher and principal each see only what they need.
Core	Reports	Daily gate summary, frequent latecomers, visitors per day, who is currently on campus.
Optional	Staff and vehicle logs	Record staff movements and vehicles entering the compound.
Optional	Parent notification	SMS or email when a child arrives late or leaves early.
Optional	Emergency roll-call mode	One-click list of everyone currently on campus.
Optional	Biometric / face recognition	Discuss privacy and data-protection issues before proposing this.

4. Project Phases and Deliverables

Phase	What you must do	Deliverables
1. Preliminary investigation	Define the problem and scope. Assess technical, economic, operational and schedule feasibility.	Problem statement; feasibility report
2. Fact-finding	Interview the gate guard and deputy principal. Observe the gate at peak times. Review existing registers. Run a short questionnaire.	Interview notes; observation log; questionnaire results
3. Requirements analysis	List functional and non-functional requirements (speed, security, usability, reliability). Prioritise them.	Requirements specification
4. System modelling	Model the current and the proposed system.	Context diagram; DFD level 0 and 1; use case diagram; flowcharts; ERD
5. System design	Design the database, forms, reports, security and backup.	Data dictionary; normalised tables; input/output layouts; interface mock-ups; system flowchart
6. Prototype	Build a working model of the key functions. Allowed tools: MS Access, Excel / Google Sheets with forms, Python, or PHP/MySQL.	Working prototype; screenshots
7. Testing	Write a test plan with test data. Record expected and actual results and fix errors.	Test plan; test results table
8. Implementation and maintenance	Plan user training, the changeover method (parallel, direct, phased or pilot) and ongoing support.	Changeover plan; user guide
9. Documentation and presentation	Compile the final report and present to the client.	Final report; 10-minute presentation

5. Guidance for Key Parts

Sample requirements (use as a model, then write your own)

- **Functional:** The system shall record the time of every entry and exit.
- **Functional:** The system shall flag any student arriving after the cut-off time.
- **Functional:** The guard shall be able to search a student by name or admission number.
- **Non-functional:** A check-in shall take less than 5 seconds.
- **Non-functional:** Only authorised users can view reports; data is backed up daily.
- **Non-functional:** The system is usable by someone with minimal computer training.

Suggested starting entities for your ERD

Entity	Suggested attributes
STUDENT	StudentID, Name, Class, ParentContact, CardNumber
STAFF	StaffID, Name, Department
VISITOR	VisitorID, Name, IDNumber, Phone
GATE_LOG	LogID, PersonType, PersonID, TimeIn, TimeOut, Reason
EXIT_PERMISSION	PermitID, StudentID, ApprovedBy, Reason, Date
USER	UserID, Role, Username, Password (stored securely)

You may add, remove or change entities if you can justify your decisions.

6. Team Roles

Each member takes one main role. Roles may be rotated between phases so everyone gains experience, but every member must contribute to the final report.

Role	Main responsibility
Project manager	Plans the work, tracks deadlines, runs team meetings, keeps a meeting log.
Systems analyst	Leads fact-finding, requirements and the process models (DFDs, use cases).
Database designer	Builds the ERD, data dictionary and normalised tables.
Interface designer	Designs forms, reports and screen mock-ups; builds the prototype interface.
Tester / documenter	Writes the test plan, records results and assembles the final report.

7. Timeline

Weeks	Phases	Checkpoint
1 - 2	Phases 1 and 2	Problem statement and fact-finding evidence shown to teacher
3 - 4	Phases 3 and 4	Requirements and draft diagrams reviewed
5 - 6	Phase 5	Database and interface design approved
7 - 8	Phases 6 and 7	Prototype demonstration and test results
9	Phases 8 and 9	Final report submitted; presentations

8. Marking Scheme (100 marks)

Component	Marks	What earns top marks
Problem definition and feasibility	10	Clear, specific problem; all four feasibility areas covered with reasons
Fact-finding evidence	15	Real interviews, observation and questionnaire with analysed results
Requirements specification	10	Complete, clear, testable and prioritised requirements
System models (DFD, ERD, use cases)	20	Correct notation, consistent across diagrams, matches requirements
Design (database, interface, security)	20	Normalised tables, usable layouts, sensible security and backup plan
Prototype and testing	15	Core features work; test plan is thorough with evidence
Documentation and presentation	10	Well organised, professional, confident and clear delivery
Total	**100**	

9. What to Submit

- One **final report** (typed) containing the deliverables for Phases 1 to 8, with a contents page.
- The **prototype** on a flash drive or shared folder, with a short user guide.
- **Evidence** of fact-finding as appendices: interview notes, questionnaire, observation log.
- A **10-minute presentation** with a 5-minute question session.
- A one-page **team contribution sheet** signed by all members.

Academic honesty

All work must be your own. Diagrams and designs must come from your own investigation of the school, not copied from other groups or the internet. Cite any source you use. Handle personal data (names, phone numbers) responsibly during fact-finding and use sample data in your prototype.

10. Final Checklist

[ ]	Problem statement and feasibility report included
[ ]	Interview notes, observation log and questionnaire attached
[ ]	Requirements are numbered and marked functional or non-functional
[ ]	Context diagram, DFD 0 and 1, use case diagram and ERD completed
[ ]	Data dictionary and normalised tables completed
[ ]	Input forms, report layouts and interface mock-ups designed
[ ]	Prototype runs and core features work
[ ]	Test plan with expected and actual results completed
[ ]	Changeover plan and user guide written
[ ]	Contribution sheet signed; presentation rehearsed

11. Extension Challenges (bonus for strong groups)

- Design an emergency roll-call screen listing everyone currently on campus.
- Propose a parent mobile app and sketch its screens.
- Write a short privacy and data-protection policy for the system.
- Show how the gate data could feed into the school's attendance system.